

Lecture 5: Applying Consumer Theory (Chapter 5 · §6.5)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, Chapter 5 — Consumer choice (applied) · §6.5 — Income and substitution effects. *Lecturer: L. Butera — deck "Lecture 5: Applying consumer theory."* (This lecture uses the machinery built in Lecture 4 — the optimum where $MRS = \text{price ratio}$ — and puts it to work.)

Résumé. Lecture 4 found the **best bundle for given prices and income**. Lecture 5 *applies* that result three ways: (1) it **derives the demand curve** — change one price, re-solve, repeat, and the optimal quantities trace out demand (closing the loop back to Lecture 2); (2) it **solves the optimum with calculus** (the substitution and Lagrangian methods) to get demand *functions* $q(p, Y)$, and introduces the **dual problem** (expenditure minimization); (3) it opens the "black box" of a price change with **income and substitution effects** — the deeper decomposition that explains *inferior* and *Giffen* goods and the puzzle of **labour supply**.

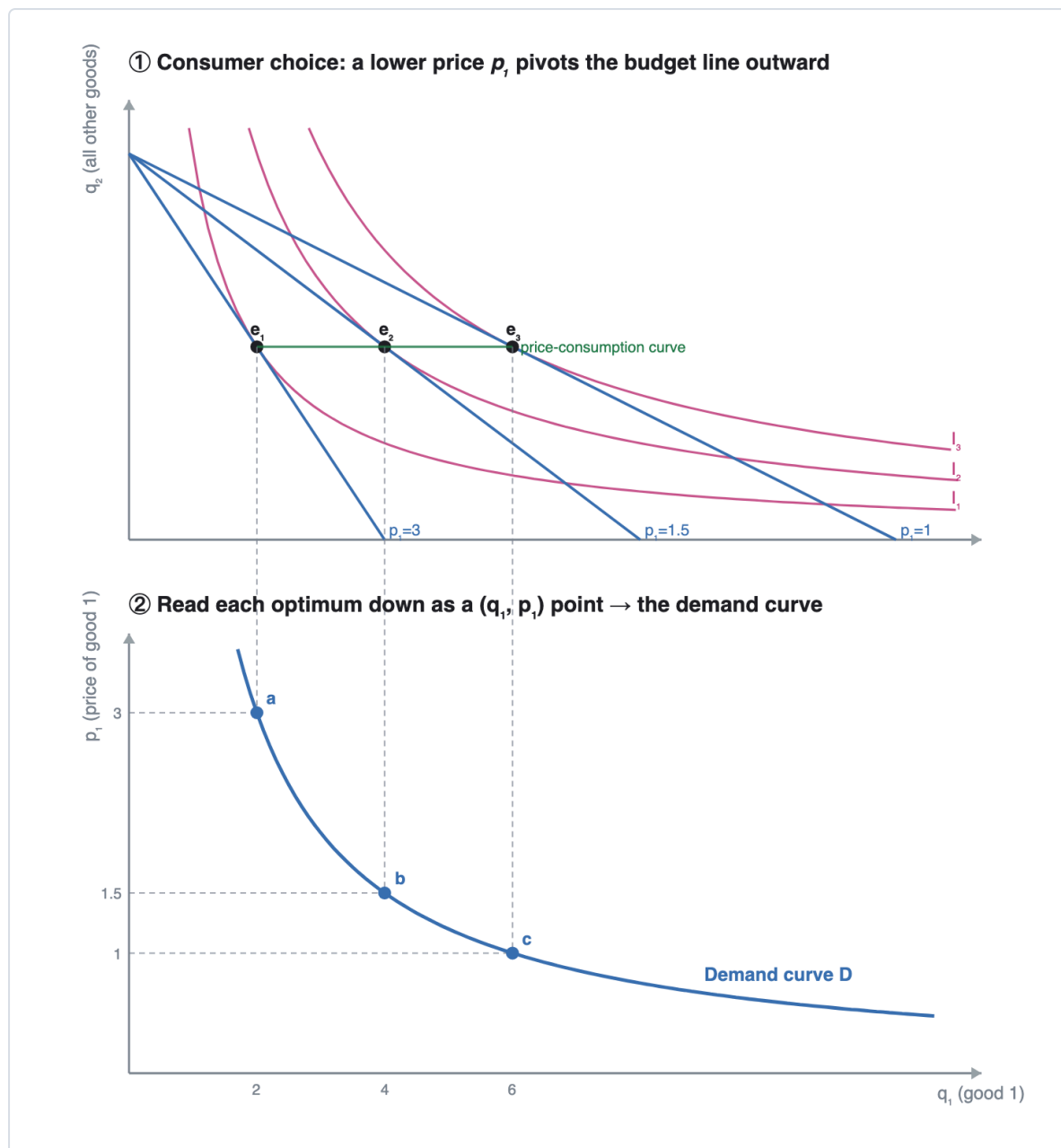
★ **The heart of the lecture:** a price change moves demand for **two** reasons, and separating them explains almost everything. The **substitution effect** (the good is now relatively dearer → switch away, **always** in the "less" direction) and the **income effect** (the price rise makes you effectively poorer → buy less if the good is **normal**, *more* if **inferior**). **Total effect = substitution effect + income effect**. When a strong, positive income effect on an inferior good *overwhelms* the substitution effect, demand slopes **up** — a **Giffen good**. The same tool cracks the wage puzzle: a raise makes you work *more* (substitution) but also lets you afford more leisure (income).

Key concepts / "modes" to use: deriving demand from the optimum · the substitution method & the Lagrangian ($MU_1/p_1 = MU_2/p_2 = \lambda$) · demand functions $q(p_1, p_2, Y)$ · the dual / expenditure minimization / Hicksian (compensated) demand · substitution effect vs income effect (Slutsky/Hicks decomposition) · normal · inferior · Giffen · the labour-leisure model & backward-bending supply.

★ Deriving the demand curve from consumer choice

Core: we never really *assumed* the demand curve of Lecture 2 — we can **derive** it. Hold **preferences, income and other prices constant**, then **change the good's own price** and re-solve the optimum. The resulting optimal quantities, plotted against price, **are** the demand curve.

- **The recipe:** at each price p_1 , find the tangency bundle ($MRS = p_1/p_2$). Lower p_1 → the budget line swings out → a **new, higher** optimal q_1 . Collect the (p_1 , q_1) pairs → the **individual demand curve** for good 1.



How to read it (building demand, step by step). ① (top) start from the consumer's optimum. Every time p_1 falls, the budget line **pivots out** along the horizontal axis (same vertical intercept Y/p_2), and the tangency slides to a **new optimum** with **more q_1** : $e_1 \rightarrow e_2 \rightarrow e_3$. ② (bottom) each optimum hands you one (**quantity, price**) pair — drop a dashed line down from e_1 to plot point $a = (q_1 \ 2, \ p_1 \ 3)$, from e_2 to $b = (4, \ 1.5)$, from e_3 to $c = (6, \ 1)$. **Join a, b, c $\rightarrow$ the demand curve D**. That's the whole trick: *the demand curve is just the consumer's optimum, re-solved and re-plotted for every price.* (This example is Cobb–Douglas $U = q_1 q_2$, so demand is $q_1^* = Y/(2p_1)$ — the smooth, unit-elastic curve shown.)

- **Movement vs shift (ties back to Ch. 2):** changing the good's **own price** traces a **movement along** its demand curve; changing **income** or **another price** re-solves to a different curve — a **shift of demand**. *In plain terms:* the demand curve is just "the optimum, redrawn for every possible own-price."
- **Why this matters:** it makes demand a *result*, not an assumption — and it means the **shape** of demand (its elasticity, whether it can slope up) is inherited from the **shape of preferences** (the indifference curves).

Solving the optimum analytically — two methods

Core: the graphical tangency $MRS = p_1/p_2$ can be solved with **calculus** to get the **demand functions** $q_1^*(p_1, p_2, Y)$ and $q_2^*(p_1, p_2, Y)$ explicitly. Two equivalent routes.

(a) The substitution method

Core: turn the constrained problem into an **unconstrained** one by substituting the budget line into utility.

1. **Budget:** $p_1 q_1 + p_2 q_2 = Y \rightarrow$ solve for one good: $q_2 = (Y - p_1 q_1)/p_2$.
2. **Substitute** into $U(q_1, q_2) \rightarrow$ utility is now a function of q_1 alone.
3. **Maximize:** set $dU/dq_1 = 0$. This reproduces the tangency condition $MU_1/MU_2 = p_1/p_2$ and, with the budget line, yields q_1^* and q_2^* as functions of (p_1, p_2, Y) .

(Requires the utility function to be **quasi-concave** — i.e. indifference curves **convex to the origin** — so the tangency is a maximum, not a minimum.)

(b) The Lagrangian method

Core: the general tool for constrained optimization (met in L4's appendix). Build a function that bolts the constraint onto the objective with a multiplier λ .

$$\mathcal{L} = U(q_1, q_2) + \lambda (Y - p_1 q_1 - p_2 q_2)$$

- **First-order conditions** (set each partial derivative to 0):
 - $\partial \mathcal{L} / \partial q_1 = MU_1 - \lambda p_1 = 0$
 - $\partial \mathcal{L} / \partial q_2 = MU_2 - \lambda p_2 = 0$
 - $\partial \mathcal{L} / \partial \lambda = Y - p_1 q_1 - p_2 q_2 = 0$ (the budget, recovered)
- **Rearrange the first two:** $MU_1/p_1 = MU_2/p_2 = \lambda$ — the "equal marginal utility per euro" rule: at the optimum the **last euro spent on each good buys the same extra utility**. Dividing them gives back $MU_1/MU_2 = p_1/p_2$ (MRS = price ratio).
- **What is λ ?** the **marginal utility of income** — how much extra utility one more euro of budget would buy. (A "shadow price" on the budget constraint.)
- Solve the three equations together $\rightarrow$ the **demand functions**. (Worked for Cobb–Douglas in the practice set — you get the tidy result that a Cobb–Douglas consumer spends **fixed budget shares** on each good.)

The dual problem — expenditure minimization

Core: utility maximization has a **mirror image**. Instead of "maximise utility for a given budget," ask "**minimise the spending needed to reach a given utility $\bar{U}$** ." Same tangency, approached from the other side.

$$\min p_1 q_1 + p_2 q_2 \quad \text{subject to} \quad U(q_1, q_2) = \bar{U}$$

- **The solution is the expenditure function $E(p_1, p_2, \bar{U})$** — the **least money** needed to achieve utility $\bar{U}$ at prices (p_1, p_2) .

- It gives the **Hicksian (compensated) demands** — quantities as functions of prices holding **utility** constant (rather than income). *This is exactly the object the **substitution effect** uses below* — it asks how you'd re-shop if you had to stay on the **same indifference curve** but faced new relative prices.
- **Duality (the payoff)**: max-utility-given-budget and min-budget-given-utility describe the **same tangency point** — two doors into the same room. The primal gives ordinary (Marshallian) demand $q(p, Y)$; the dual gives compensated (Hicksian) demand $q(p, \bar{U})$.

★ Income and substitution effects (§6.5)

Core: the deeper "behind-the-scenes" reading of a price change. When p_1 rises, quantity demanded moves for **two distinct reasons**, and the total is their sum:

$$\text{Total effect} = \text{substitution effect} + \text{income effect}$$

- **Substitution effect** — the change in quantity from the shift in **relative prices alone**, holding utility constant. If p_1 rises, good 1 is now **relatively dearer**, so you switch toward the (now cheaper) other good. **Always negative (or zero)**: a price rise $\Rightarrow$ substitution away. *(Formally, this is the move along the **original indifference curve** — the Hicksian/compensated response from the dual.)*
- **Income effect** — the change in quantity because the price rise makes you **effectively poorer** (your income now buys less). Sign depends on the good:
 - **normal good** $\rightarrow$ you buy **less** when poorer $\Rightarrow$ income effect is **negative** (reinforces substitution);
 - **inferior good** $\rightarrow$ you buy **more** when poorer $\Rightarrow$ income effect is **positive** (opposes substitution).

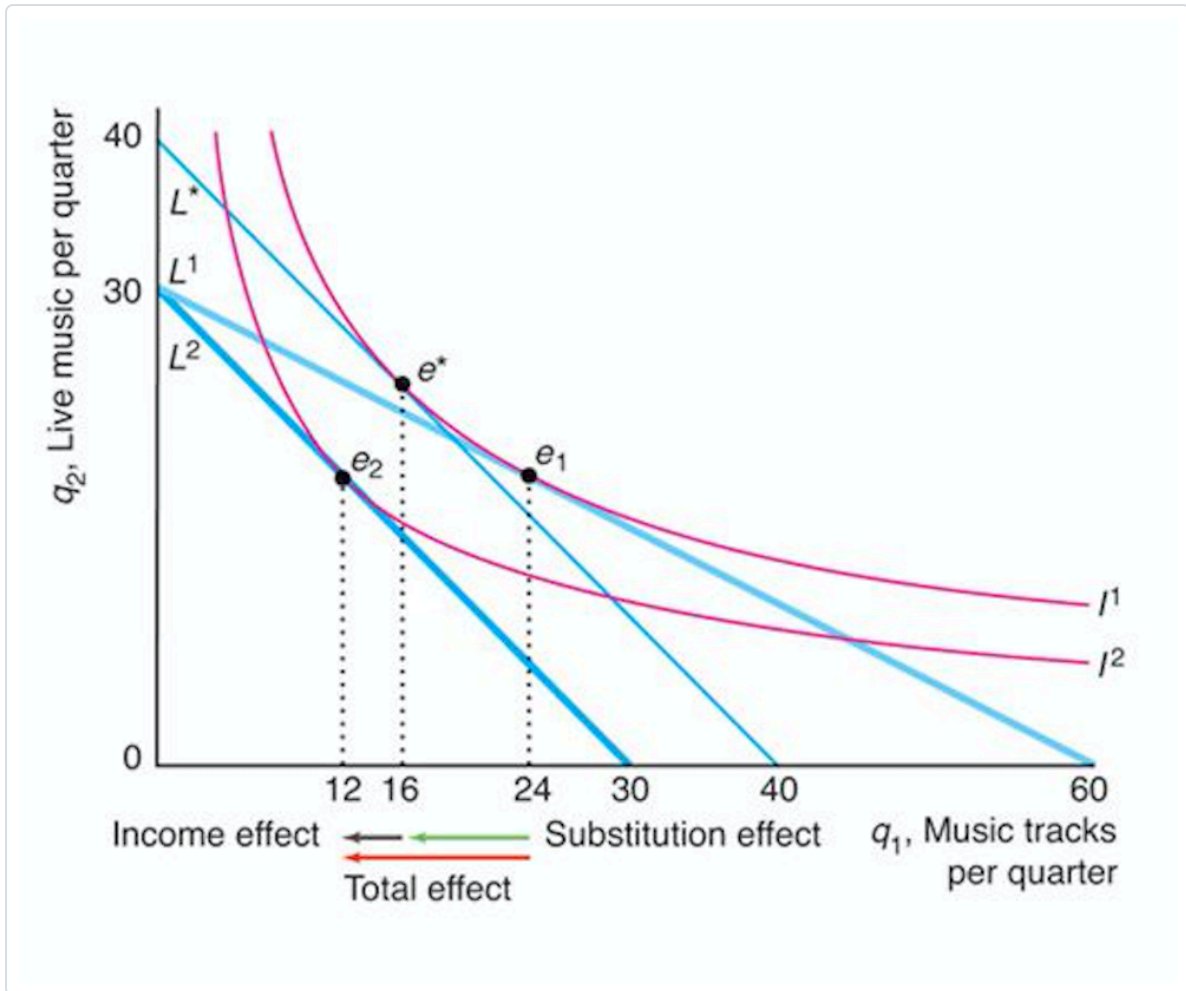
How the decomposition works (the three-step graph)

Core: compensate, then take the compensation away.

1. **Start** at optimum e^* on indifference curve I^1 with budget line L^1 .
2. **Raise p_1** $\rightarrow$ budget line pivots inward and **steeper** (L^2) $\rightarrow$ new optimum e_2 on a lower curve I^2 .
The move $e^* \rightarrow e_2$ is the **total effect**.
3. **Isolate substitution**: slide a budget line with the **new (steeper) slope** out until it is **tangent to the original curve I^1** (this " L^* " imagines you were **compensated** to stay as happy as before). Its tangency e_1 gives the **substitution effect** $e^* \rightarrow e_1$ (a slide *along* I^1).
4. **Income effect** is the leftover **parallel shift** $L^* \rightarrow L^2$, i.e. $e_1 \rightarrow e_2$ — the pure loss of purchasing power.

A normal good — the two effects reinforce

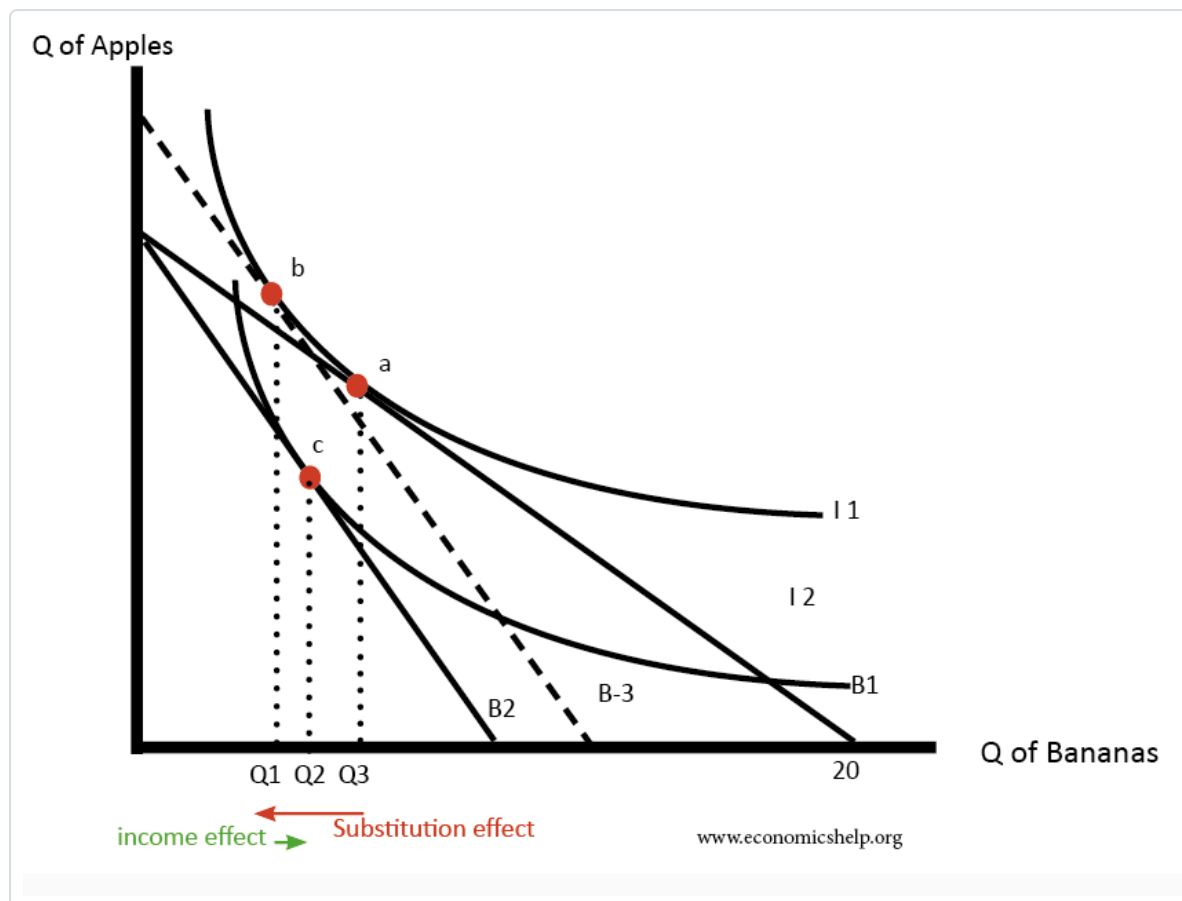
Core: for a **normal good** both effects push the **same way**, so demand falls **noticeably** when price rises.



How to read it. e^* is the original optimum. When the price of music tracks rises, the budget line pivots to L^2 and the optimum falls to e_2 . The compensated line L^* (new slope, original indifference curve I^1) locates e_1 : *substitution effect* = $e \rightarrow e_1^*$ ($24 \rightarrow 16$), *income effect* = $e_1 \rightarrow e_2$ ($16 \rightarrow 12$). For a normal good the income arrow points the **same way** as the substitution arrow $\rightarrow$ the two **add up** to a large total fall ($24 \rightarrow 12$).

An inferior good — the effects oppose

Core: for an **inferior** good the positive income effect **partly cancels** the substitution effect, so demand is **less responsive** to price.



How to read it. The substitution effect still pushes the good **down** ($a \rightarrow b$), but because the good is **inferior**, being poorer makes you want **more** of it — the income arrow points **back the other way**, offsetting part of the substitution move. **Net:** demand still falls, but by **less** than for a normal good. "The more inferior the good (the more negative its income elasticity) and the fewer the substitutes, the less demand shrinks when price rises."

Giffen goods — when demand slopes up

Core: the rare, dramatic case. A **Giffen good** is an **inferior** good whose **income effect** is so **strong** it **dominates** the substitution effect → **quantity demanded rises when price rises** → an **upward-sloping** demand curve, violating the law of demand.

- **The recipe for a Giffen good:** (1) strongly **inferior**; (2) takes up a **large share** of the budget; (3) has **few substitutes**. Then a price rise makes the household so much poorer that it **cuts the "luxury"** it used to afford and buys **even more** of the cheap staple.
- **The classic examples:** **potatoes in Ireland** in the 1840s (the origin of the name — still contested) and, more convincingly, **rice among poor households in China** (Jensen & Miller, 2008): when rice got dearer, families could no longer afford *any* meat, so they bought **more** rice to fill up. → "They can only afford rice."
- **Why it's the exception, not the rule:** it needs a strong income effect (big budget share) *and* strong inferiority — rare in rich economies. The substitution effect usually wins, which is why demand almost always slopes down.

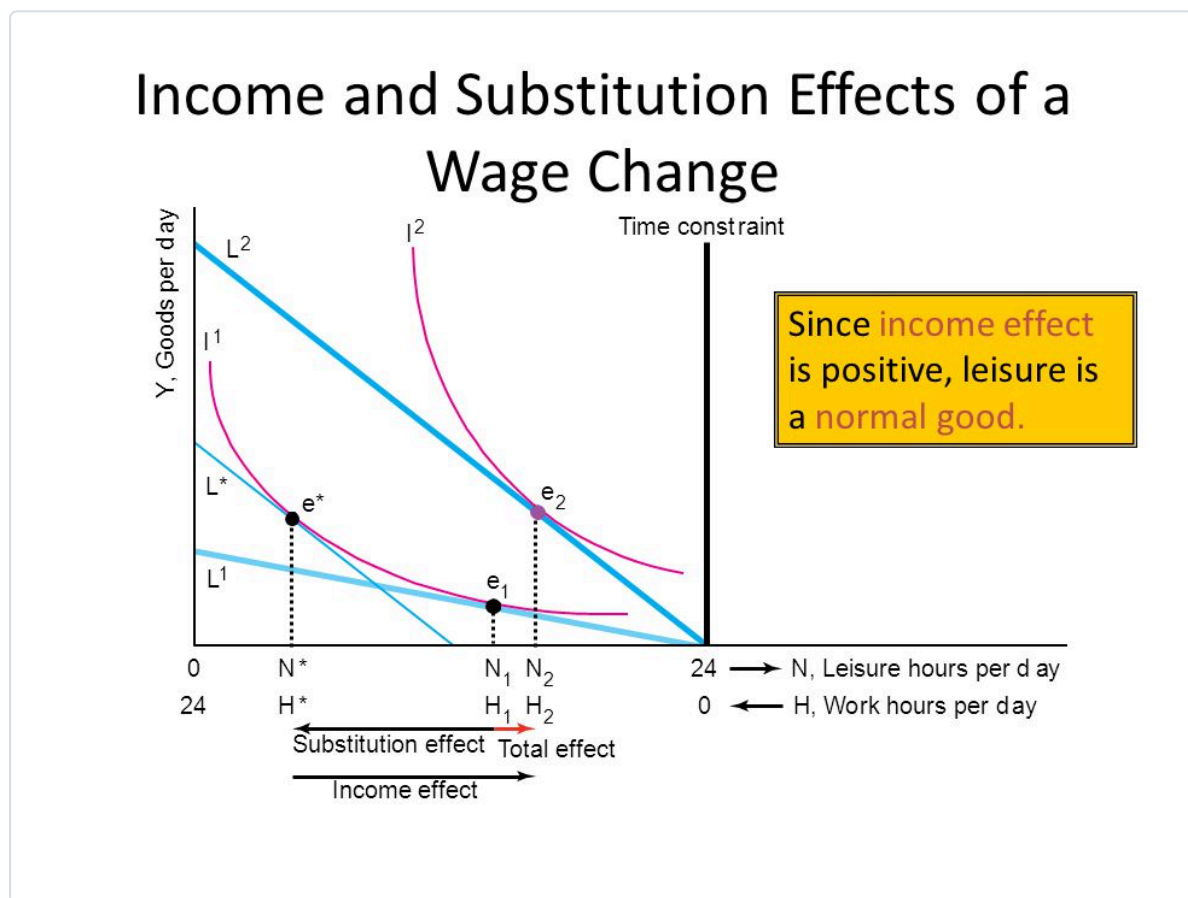
(Aside — **Hicks vs Slutsky:** two ways to "compensate" for the substitution effect. **Hicks** keeps you on the **original indifference curve** (same utility); **Slutsky** keeps the **original basket just affordable** (rotate the budget line around

the starting point). Hicks is the standard version; both give the same intuition.)

★ Application: labour supply (the wage puzzle)

Core: the decomposition cracks a real puzzle — *does a pay rise make you work more or less?* Model the worker choosing between **leisure** and **consumption (income)**. A wage is the **price of leisure** (an hour off costs you one hour's wage). So a **wage rise = leisure becomes more expensive** → apply income & substitution effects to *leisure*.

- **Substitution effect:** leisure is now dearer → consume less leisure → **work more**.
- **Income effect:** a higher wage makes you **richer**; if leisure is a **normal good**, you want **more** of it → **work less**.
- **Net effect is ambiguous** — it depends on which dominates.



How to read it. The horizontal axis is **leisure hours N** (read right-to-left, it's **work hours H**); the vertical axis is **goods/income**. A wage change tilts the budget line. Splitting $e^* \rightarrow e_2$ at the compensated point e_1 : the **substitution effect** cuts leisure (more work), the **income effect** adds leisure (less work — "since the income effect is positive, leisure is a normal good"). Because they pull **opposite ways**, the **backward-bending labour-supply curve** becomes possible: at **low** wages the substitution effect usually wins (higher wage → work more); at **high** wages the income effect can win (higher wage → work less).

- **The deck's puzzle made concrete:** a raise from 200 → 300 DKK/hr probably makes you **work more** (substitution dominates); a raise from 200 → 4000 DKK/hr might make you **work less** — you're now rich enough to "buy back" your time (income dominates).

Formula sheet (Lecture 5)

Tool	Formula
Tangency (optimum)	$MRS = MU_1/MU_2 = p_1/p_2$
Budget constraint	$p_1q_1 + p_2q_2 = Y$
Lagrangian	$\mathcal{L} = U(q_1, q_2) + \lambda(Y - p_1q_1 - p_2q_2)$
Lagrangian FOCs	$MU_1 = \lambda p_1, MU_2 = \lambda p_2 \rightarrow MU_1/p_1 = MU_2/p_2 = \lambda$
λ interpretation	marginal utility of income (shadow price of the budget)
Expenditure function (dual)	$E(p_1, p_2, \bar{U}) = \text{min spending to reach utility } \bar{U}$
Slutsky decomposition	total effect = substitution effect + income effect
Substitution effect	≤ 0 always (holding utility constant)
Income effect (price $\uparrow$)	normal: negative · inferior: positive · Giffen: positive & dominant
Cobb–Douglas $U = q_1^a q_2^b$ demand	$q_1^* = a/(a+b) \cdot Y/p_1$ (constant budget shares)

★ Practice problems (with solutions)

Do these — Lecture 5 is where the calculus of consumer choice actually gets tested. Solutions below.

1. A consumer has $U(q_1, q_2) = q_1 q_2$, income Y , prices p_1, p_2 . Use the **substitution method** to find the demand for q_1 .
2. Same consumer, now use the **Lagrangian** method to find q_1^*, q_2^* , and interpret λ .
3. Show that with **Cobb–Douglas** $U = q_1^a q_2^b$, the consumer spends a **constant share** of income on each good. What is the own-price elasticity of the demand $q_1^* = [a/(a+b)] \cdot (Y/p_1)$?
4. The price of good 1 (a **normal** good) rises. On an indifference-curve diagram, identify the **substitution** and **income** effects, and say which direction each pushes q_1 .
5. Classify the income effect (sign, for a **price rise**) and say what happens to total demand: (a) a normal good; (b) an inferior good; (c) a Giffen good.
6. Your hourly wage doubles. Using income & substitution effects, explain why you might **work more or less**. Which effect is the "substitution" one?
7. (Concept.) What does the **expenditure function** $E(p_1, p_2, \bar{U})$ measure, and how does it relate to the **substitution effect**?

Solutions

1. Budget: $q_2 = (Y - p_1 q_1)/p_2$. Substitute: $U = q_1 \cdot (Y - p_1 q_1)/p_2$. Differentiate wrt q_1 and set to 0: $dU/dq_1 = (Y - 2p_1 q_1)/p_2 = 0 \rightarrow Y - 2p_1 q_1 = 0 \rightarrow$

$$q_1^* = Y / (2p_1)$$

(and by symmetry $q_2^* = Y/(2p_2)$). The consumer spends **half** the budget on each good.

2. $\mathcal{L} = q_1 q_2 + \lambda(Y - p_1 q_1 - p_2 q_2)$. FOCs: $q_2 = \lambda p_1$ and $q_1 = \lambda p_2$ and the budget. Divide the first two: $q_2/q_1 = p_1/p_2 \rightarrow p_1 q_1 = p_2 q_2$. Put into the budget $p_1 q_1 + p_2 q_2 = Y \rightarrow 2p_1 q_1 = Y \rightarrow q_1^* = Y/(2p_1)$, $q_2^* = Y/(2p_2)$ (same as Q1 ✓). $\lambda = q_2/p_1 = Y/(2p_1 p_2)$ — the **marginal utility of an extra euro** of income.

3. Lagrangian FOCs give $MU_1/MU_2 = (a q_1^{a-1} q_2^b)/(b q_1^a q_2^{b-1}) = (a/b)(q_2/q_1) = p_1/p_2 \rightarrow p_1 q_1 = (a/b)p_2 q_2$. Sub into the budget $\rightarrow p_1 q_1 = [a/(a+b)] \cdot Y$. So **spending on good 1 is the fixed share $a/(a+b)$ of income** (independent of prices) $\rightarrow q_1^* = [a/(a+b)] \cdot (Y/p_1)$. **Own-price elasticity** = $(dq_1/dp_1)(p_1/q_1) = -[a/(a+b)]Y \cdot p_1^{-2} \cdot p_1/q_1 = -1$: Cobb–Douglas demand is **unit-elastic** (which is exactly why expenditure share is constant — recall [Lecture 3](#): unit elasticity $\Rightarrow$ spending unchanged).

4. Original optimum e^* on I^1 . Price $\uparrow \rightarrow$ new optimum e_2 on a lower I^2 (total effect, q_1 falls). Compensated line (new slope, tangent to I^1) gives e_1 : **substitution effect $e^* \rightarrow e_1$** pushes q_1 **down** (dearer relative price); **income effect $e_1 \rightarrow e_2$** also pushes q_1 **down** (poorer, and it's normal). Both **negative** $\rightarrow$ they **reinforce**. (*This is the normal-good figure.*)

5. For a **price rise**: (a) **normal** $\rightarrow$ income effect **negative**; total demand **falls** (both effects down). (b) **inferior** $\rightarrow$ income effect **positive**, partly offsets substitution; total demand **falls, but less**. (c) **Giffen** $\rightarrow$ income effect **positive and dominant**; total demand **rises** (law of demand violated).

6. A wage rise makes leisure **more expensive** (its price = the wage). **Substitution effect**: leisure dearer $\rightarrow$ take less leisure $\rightarrow$ **work more**. **Income effect**: you're richer and leisure is normal $\rightarrow$ want more leisure $\rightarrow$ **work less**. The **substitution effect** is the "leisure-is-now-dearer" one; whether you net work more or less depends on which dominates ($\rightarrow$ backward-bending supply).

7. $E(p_1, p_2, \bar{U})$ = the **minimum spending needed to reach utility $\bar{U}$** at given prices (the dual). Its quantities are the **Hicksian/compensated demands** — precisely what the **substitution effect** uses, since that effect asks how you re-optimize when **utility is held fixed** and only **relative prices** change.

★ Fun facts & memorable details (Lecture 5)

Sticky bits from applying consumer theory.

- **The demand curve is a result, not an assumption.** Wiggle one price, re-solve the optimum over and over, and demand **draws itself**. Its whole shape is inherited from the curvature of your indifference curves.

- **λ has a name: the marginal utility of money.** The Lagrange multiplier isn't just algebra — it tells you **how much happier one extra euro would make you**.
 - **Giffen goods really exist — barely.** For 150 years the **Irish-potato** story was economists' only (contested) example. Then **Jensen & Miller (2008)** found it for real: subsidise rice for poor Chinese families and they bought *less* — because they could finally afford some meat.
 - **"They can only afford rice."** The heart of the Giffen paradox: when your staple gets pricier, you're too poor to buy anything *but* the staple — so you buy **more** of it.
 - **A pay rise can make you work less.** The **backward-bending labour supply curve**: past some wage, you're rich enough to "buy back" your own time. Lottery winners who quit their jobs are the income effect made visible.
 - **Every price change is secretly two changes.** Splitting the substitution effect from the income effect is one of the most useful moves in all of microeconomics — it's why we can tell a normal good from an inferior one from a Giffen good just by signing an arrow.
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